Zazr and VAVR Tour

Replace null-config lookup with Option
Season 0 · Libraries · Beginner

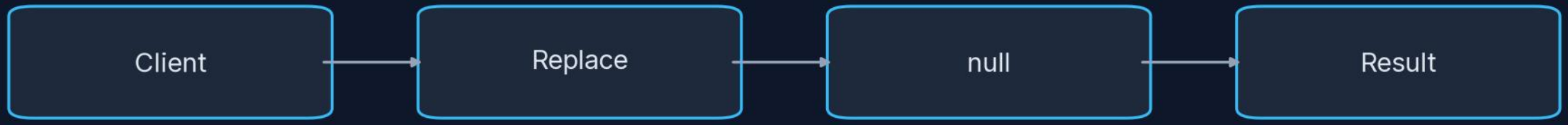
Today's Hook

- Problem: Replace null-config lookup with Option
- Naive pain cue: none
- We will refactor to Zazr & VAVR with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Replace null-config lookup with Option
- Constraints: solve «Replace null-config lookup with Option» with Zazr & VAVR — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Replace null-config lookup with Option
- Success looks like: none

Structure



Zazr & VAVR: Replace null-config lookup with Option

Real-World Scenario

- Running example: Replace null-config lookup with Option
- Key types: OptionCompare
- Happy-path stdout cue: none
- Failure-path stdout cue: none

Naive Code

```
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;

import java.util.*;

public final class NaiveOpt {
  public static String describe(String s){ return s==null? "none": "some:"+s; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Replace null-config lookup with Option
- Without Zazr & VAVR, call sites absorb the wrong details.
- Failure mode to show on camera: none
- New requirements force shotgun edits across the naive code.

Solution

```java
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;

/**
 * Side-by-side: same Option describe / map+flatMap program on real VAVR and real Zazr.
 */
public final class OptionCompare {
    private OptionCompare() {}

    public static String vavrDescribe(String raw) {
        io.vavr.control.Option<String> o = raw == null
                ? io.vavr.control.Option.none()
                : io.vavr.control.Option.some(raw);
        return o.map(s -> "vavr-some:" + s).getOrElse("vavr-none");
    }

    public static String zazrDescribe(String raw) {
        dev.zazr.control.Option<String> o = raw == null
                ? dev.zazr.control.Option.none()
                : dev.zazr.control.Option.some(raw);
        return o.map(s -> "zazr-some:" + s).getOrElse("zazr-none");
    }
```

Solution (2)

```
    /** Trim → reject blank → lowercase — identical pipeline on both libraries. */
    public static String vavrMapFlat(String raw) {
        return io.vavr.control.Option.of(raw)
                .flatMap(s -> s.isBlank() ? io.vavr.control.Option.none() : io.vavr.control.Option.some(s.trim()))
                .map(String::toLowerCase)
                .map(s -> "vavr-some:" + s)
                .getOrElse("vavr-none");
    }

    public static String zazrMapFlat(String raw) {
        return (raw == null ? dev.zazr.control.Option.<String>none() : dev.zazr.control.Option.some(raw))
                .flatMap(s -> s.isBlank() ? dev.zazr.control.Option.none() : dev.zazr.control.Option.some(s.trim()))
                .map(String::toLowerCase)
                .map(s -> "zazr-some:" + s)
                .getOrElse("zazr-none");
    }
}
```

Runnable main

```java
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;

/** Problem: same Option describe — real io.vavr vs real dev.zazr. */
public final class ProblemDemo {
    public static void main(String[] args) {
        System.out.println("=== VAVR vs Zazr: real Option APIs ===");
        System.out.println("--- naive null String ---");
        System.out.println(NaiveOpt.describe(null));
        System.out.println(NaiveOpt.describe("Hi"));

        System.out.println("--- describe (some/none) ---");
        System.out.println(OptionCompare.vavrDescribe("x"));
        System.out.println(OptionCompare.zazrDescribe("x"));
        System.out.println(OptionCompare.vavrDescribe(null));
        System.out.println(OptionCompare.zazrDescribe(null));

        System.out.println("--- map/flatMap pipeline ---");
        System.out.println(OptionCompare.vavrMapFlat("  Hello "));
        System.out.println(OptionCompare.zazrMapFlat("  Hello "));
        System.out.println(OptionCompare.vavrMapFlat("   "));
        System.out.println(OptionCompare.zazrMapFlat("   "));
        System.out.println(OptionCompare.vavrMapFlat(null));
        System.out.println(OptionCompare.zazrMapFlat(null));
    }
```

Runnable main (2)

```
}
```

Walkthrough Output

```
=== VAVR vs Zazr: same Option describe ===
--- naive ---
none
--- compare ---
zazr-some:x
vavr-some:x
zazr-none
vavr-none
```

Step-by-Step Trace

- Match stdout to stages — look for: none
- Then the injected failure: none
- Re-run: java -cp ... 02-zazr-and-vavr ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```java
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    }

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        return o.map(s -> "zazr-some:" + s).getOrElse("zazr-none");
    }
```

Modern Java 25 / Zazr (2)

```
/** Trim → reject blank → lowercase — identical pipeline on both libraries. */
public static String vavrMapFlat(String raw) {
    return io.vavr.control.Option.of(raw)
            .flatMap(s -> s.isBlank() ? io.vavr.control.Option.none() : io.vavr.control.Option.some(s.trim()))
            .map(String::toLowerCase)
            .map(s -> "vavr-some:" + s)
            .get
```

Trade-offs

- Use Zazr & VAVR when «Replace null-config lookup with Option» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Zazr & VAVR in product language.

Related Patterns

- Name the pattern usually confused with Zazr & VAVR and separate the forces.
- Sketch the roles for «Replace null-config lookup with Option» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Replace null-config lookup with Option
- Naive pain → Zazr & VAVR boundary → Demo stdout.
- Remember the failure cue: none
- Break a ProblemTest assertion, watch it fail, restore it.